



Climate change and extreme weather events

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Unseen Signals: How Climate Reshapes Life



Climate is changing
not a debate
anymore.



Extremes are
intensifying.



Biodiversity:
impacted in hidden,
complex ways.



The first cracks:
physiological level.

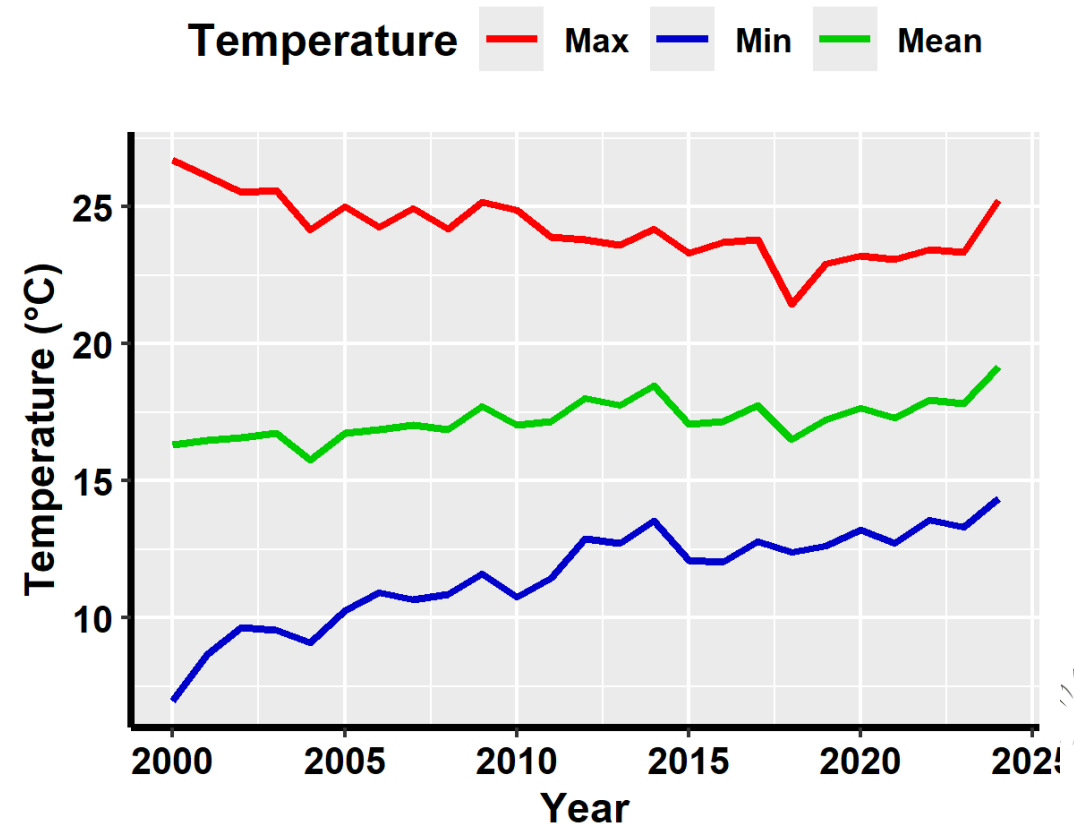
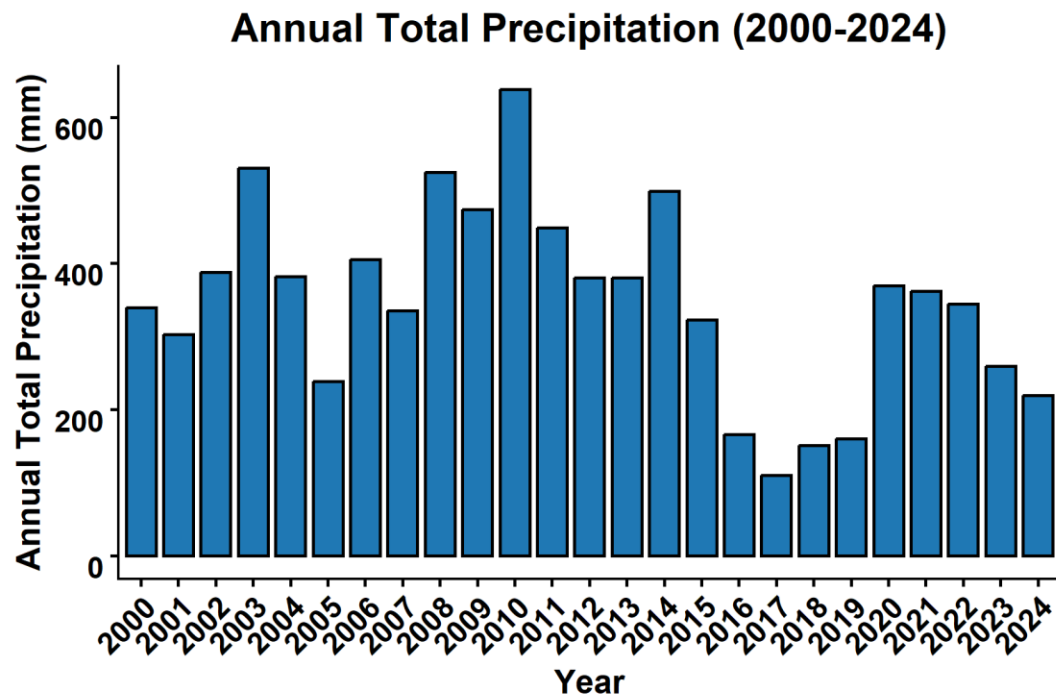


Signals:
subtle & invisible.



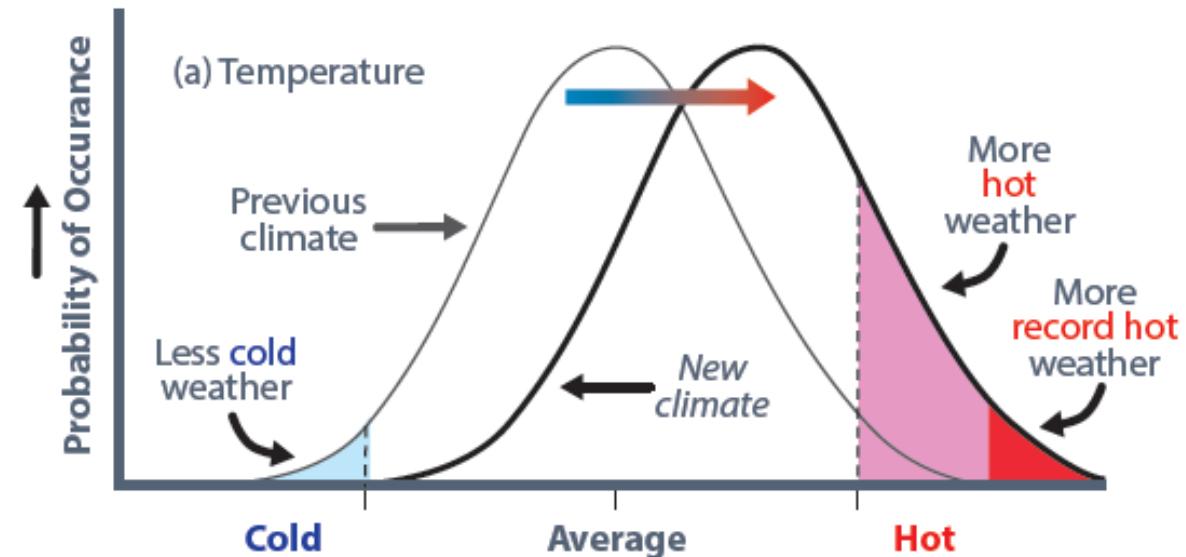
Global Warming vs. Climate Variability

- Climate change is often simplified as a steady rise in global mean temperature.
- However, it manifests as **increased variability** and **extreme anomalies**.



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- Climate change is often simplified as a steady rise in global mean temperature.
- However, it manifests as **increased variability** and **extreme anomalies**.
- The Baseline Shift: As the average global temperature rises, the entire "bell curve" of weather shifts.
- what used to be a "rare" hot day now falls into the "frequent" category, and "impossible" heat becomes the new extreme.



Lloyd and George (2022). *Proceedings of the Royal Society of Queensland*

Climate Change is no longer just a trend in average global temperatures
➔ **fundamental shift in the frequency and intensity of weather anomalies.**



Climate Change and Extreme Weather Events

Cold Snaps



Heavy Rainfall

Drought

Heat Waves



“A period of abnormally hot weather, often defined with reference to a relative temperature threshold, lasting from two days to months.”

Heatwaves: the Threshold Challenge

- Absolute thresholds (e.g., $T_{\max} \geq 35^{\circ}\text{C}$ for $\geq N$ days)
- *no single global definition, as $>30^{\circ}\text{C}$ is a heatwave in Scandinavia but a normal day in the Sahara.*
- Impact-based metrics (heat index / wet-bulb / Excess Heat Factor).
- Compound definitions (day + night heat: $T_{\max}$ & $T_{\min}$).



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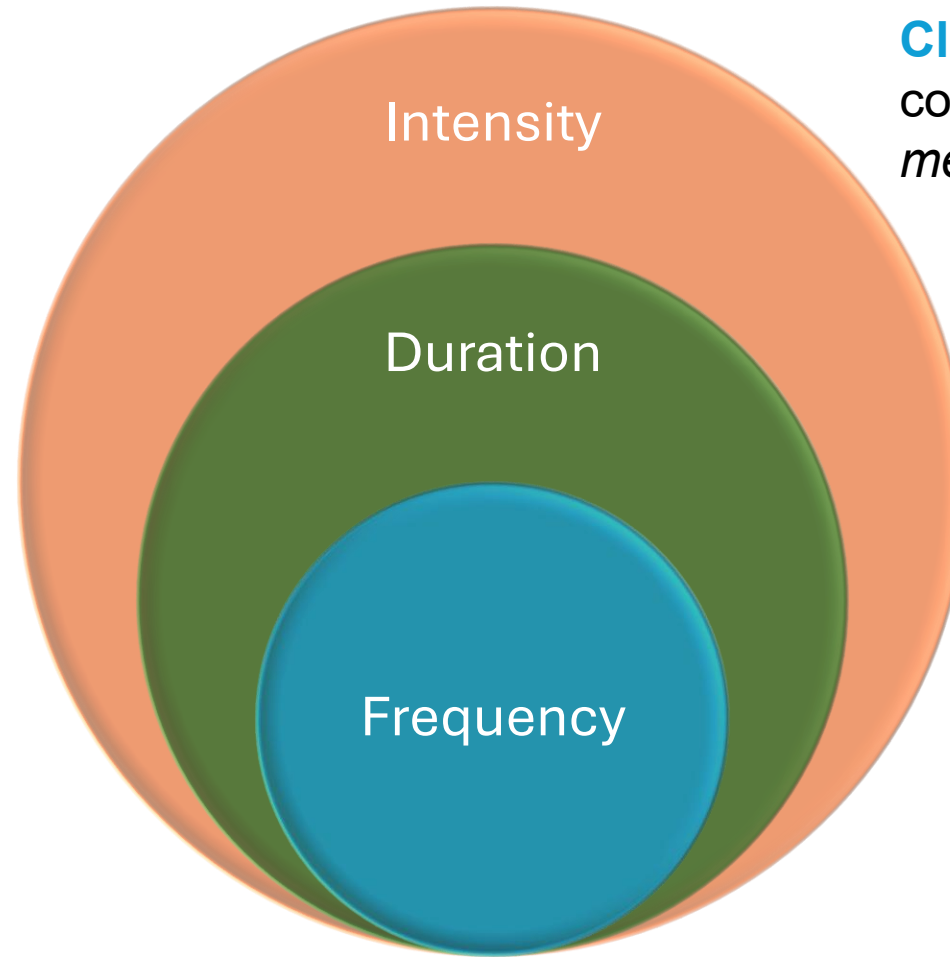
Extreme Rainfall: The Return Period

- **Return period / “X-year event”:** 1-in-100 Year Event and refers to an event that has a 1% probability of occurring in any given year.
- Fixed absolute thresholds.
- **Block maxima indices:**
 - **Rx1day** = maximum 1-day precipitation in a period.
 - **Rx5day** = maximum consecutive 5-day precipitation in a period.

How severe and sustained

Heat Waves

Temperatures exceeding
“normal values” over a
minimum period of time.
(Tmax/Tmin and N)



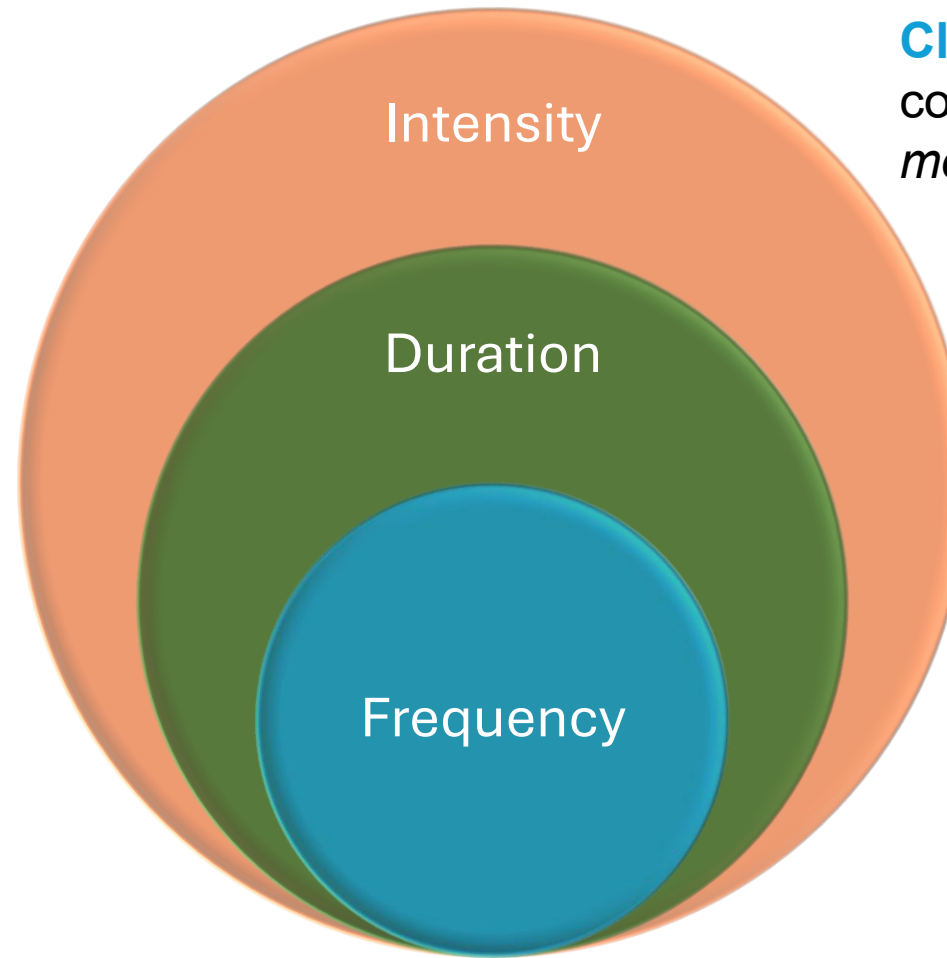
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Anomaly Detection: Highlights deviations above the norm using percentiles: *above 90-95% of the observed daily temperatures*



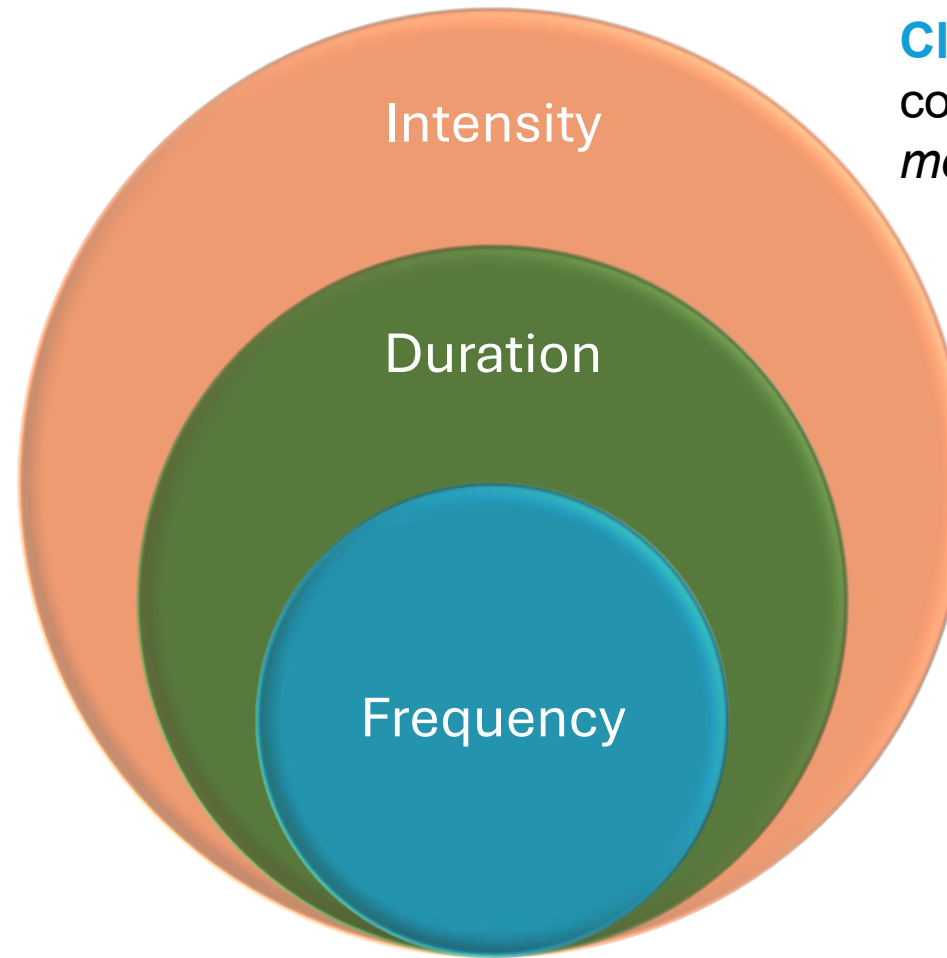
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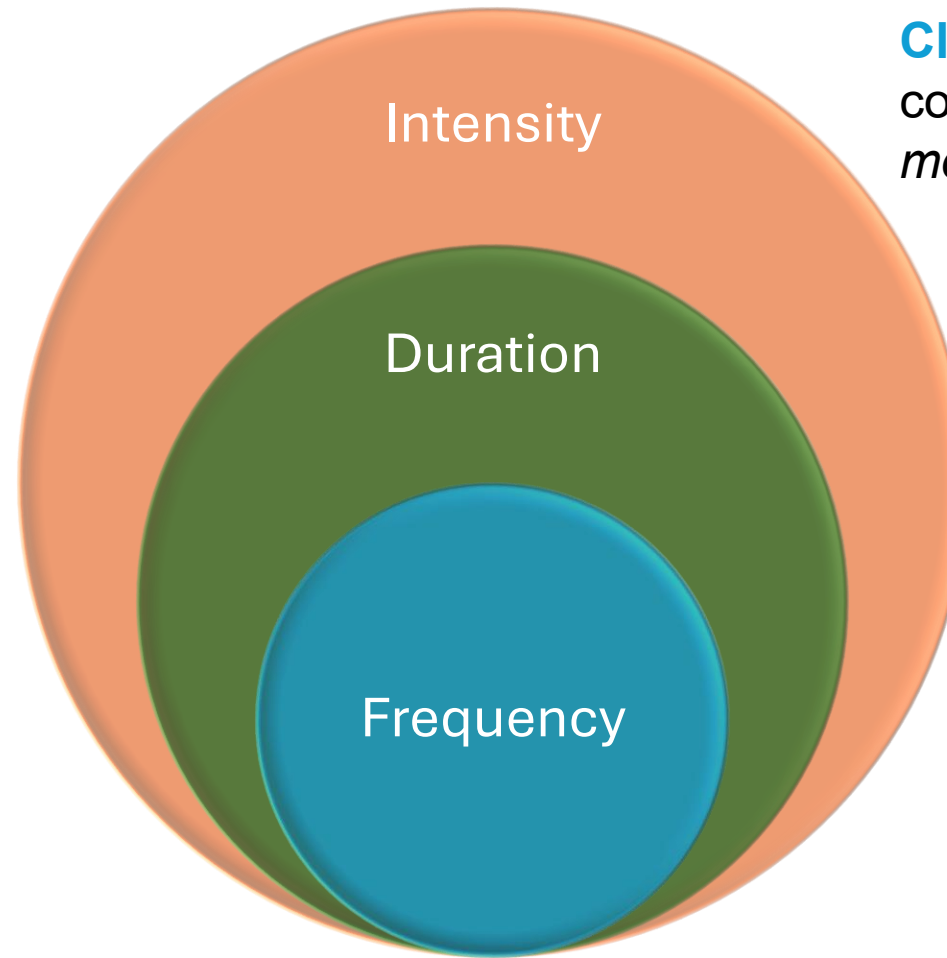
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Climate Change and Extreme Weather Events

Heatwaves
heatwaveR (Hobday et al.
method)

Goal: Detect “heatwave events” as **periods of unusually high temperature** relative to a seasonal baseline

- **Input needed:** A (near) **continuous daily** time series with
 - t = Date (YYYY-MM-DD)
 - temp = your metric (e.g., **Tmax**, **Tmean**, etc.)



Climate Change and Extreme Weather Events

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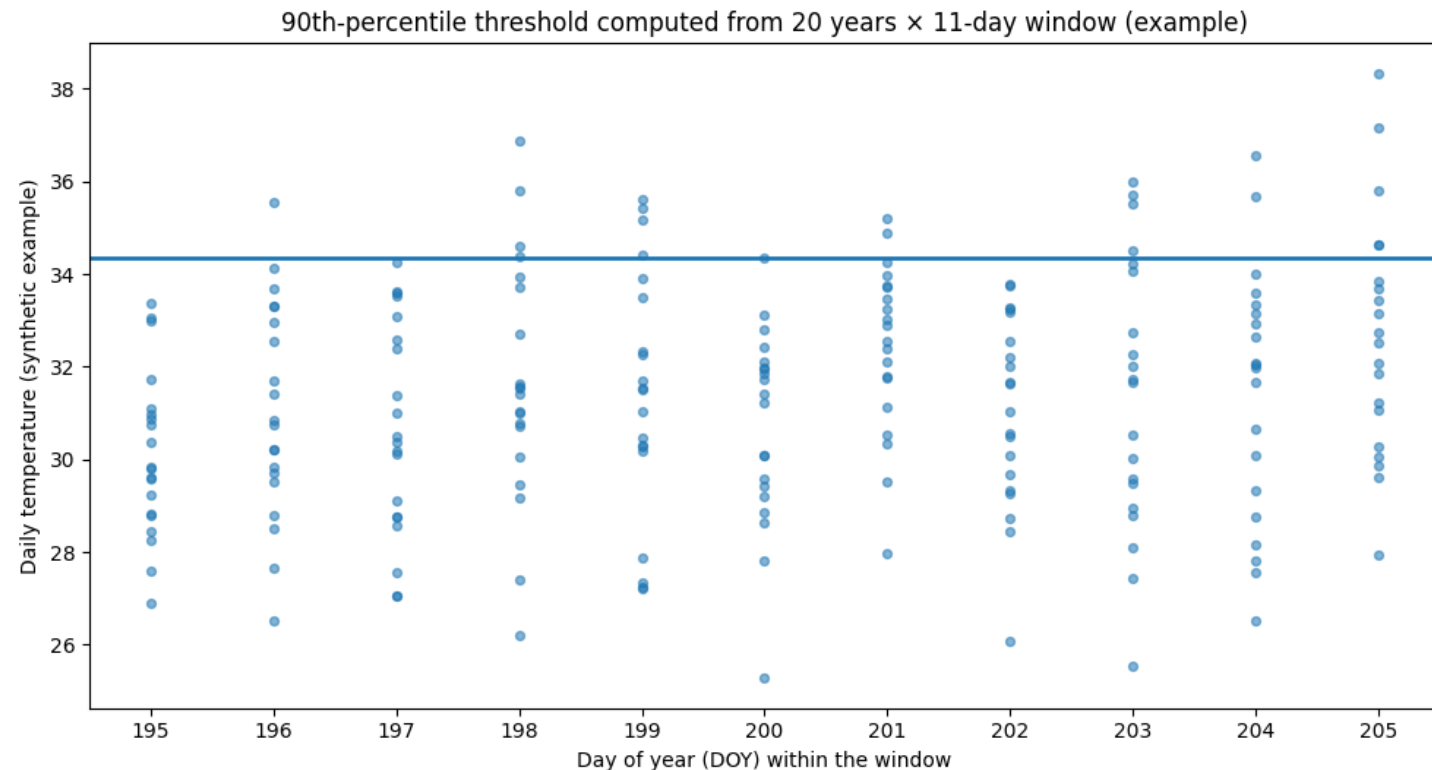
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- **Step 1: Build climatology & threshold (ts2clm)**

- Choose a **climatology period**
- Choose a threshold (default **90th percentile**)
- How to compute the threshold?

It pools data in a **sliding window around each day-of-year** with default `windowHalfWidth = 5` → an **11-day window** (± 5 days).



Climate Change and Extreme Weather Events

Heatwaves heatwaveR (Hobday et al. method)

- **Step 2: Detect events (detect_event)**
 - A heatwave is detected when **temp > thresh** for at least **minDuration = 5 days** (default).

Outputs:

- **Climatology table:** full time series with seas, thresh and event flags.
- **Event table:** start/end dates, peak date, duration, intensity metrics (mean/max/cumulative)

```
$climatology
# A tibble: 5,744 × 9
```

	doy	t	temp	seas	thresh	threshCriterion	duration	durationCriterion	event	event_no
	<int>	<date>	<dbl>	<dbl>	<dbl>	<lgl>	<lgl>	<lgl>	<lgl>	<int>
1	1	2010-01-01	12.7	13.8	17.9	FALSE	FALSE	FALSE	FALSE	NA
2	2	2010-01-02	17.0	13.8	17.9	FALSE	FALSE	FALSE	FALSE	NA
3	3	2010-01-03	18.3	13.8	17.8	TRUE	FALSE	FALSE	FALSE	NA
4	4	2010-01-04	16.1	13.7	17.7	FALSE	FALSE	FALSE	FALSE	NA
5	5	2010-01-05	10.9	13.7	17.7	FALSE	FALSE	FALSE	FALSE	NA
6	6	2010-01-06	12.5	13.6	17.6	FALSE	FALSE	FALSE	FALSE	NA
7	7	2010-01-07	10.1	13.6	17.6	FALSE	FALSE	FALSE	FALSE	NA
8	8	2010-01-08	4.59	13.5	17.6	FALSE	FALSE	FALSE	FALSE	NA
9	9	2010-01-09	6.64	13.5	17.6	FALSE	FALSE	FALSE	FALSE	NA
10	10	2010-01-10	11.7	13.4	17.6	FALSE	FALSE	FALSE	FALSE	NA

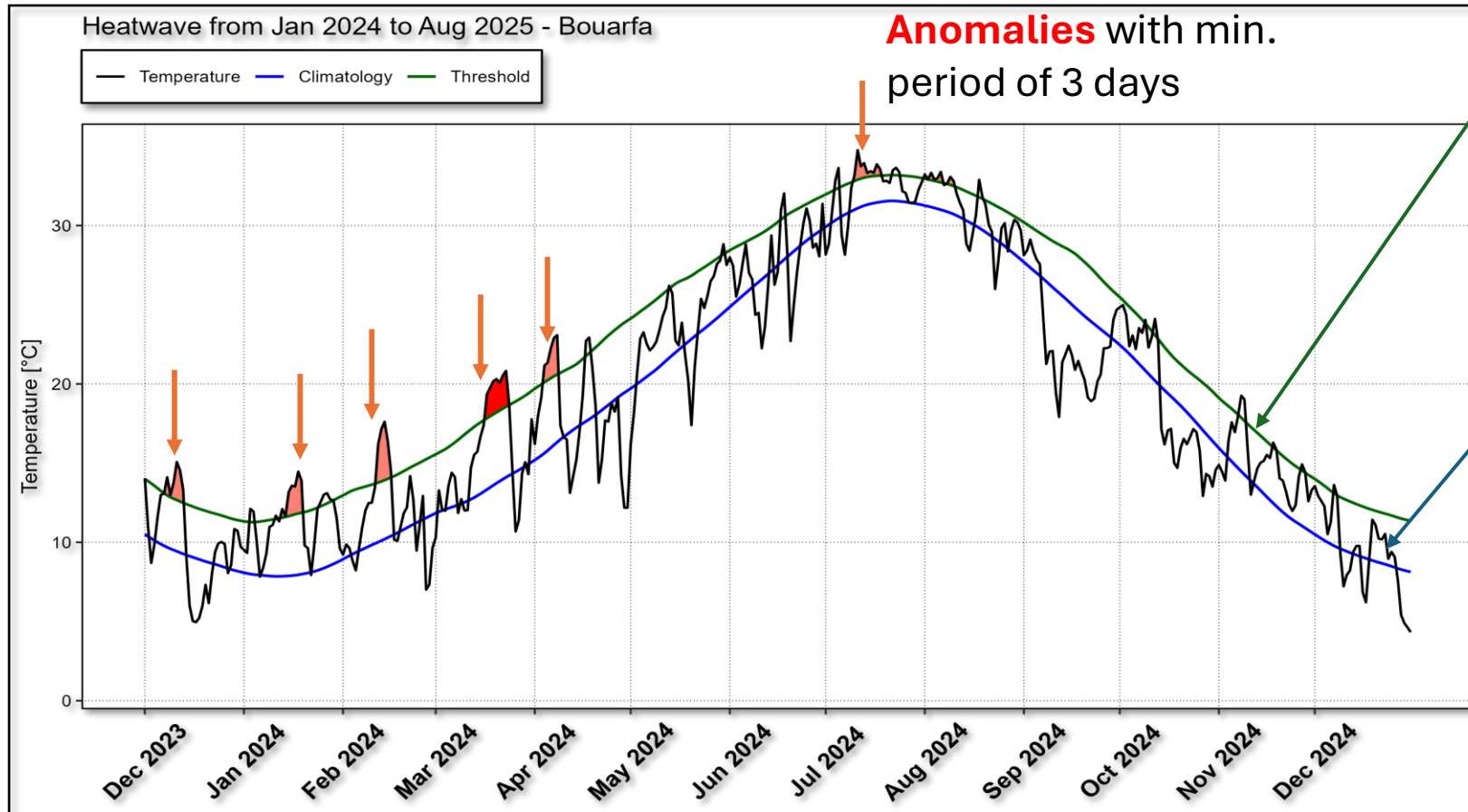
```
$event
# A tibble: 28 × 22
```

	event_no	index_start	index_peak	index_end	duration	date_start	date_peak	date_end	intensity_mean	intensity_max
	<int>	<int>	<int>	<int>	<dbl>	<date>	<date>	<date>	<dbl>	<dbl>
1	1	1083	1088	1089	7	2012-12-18	2012-12-23	2012-12-24	5.25	6.57
2	2	1955	1957	1961	7	2015-05-09	2015-05-11	2015-05-15	7.77	9.13
3	3	2191	2192	2195	5	2015-12-31	2016-01-01	2016-01-04	5.10	5.58
4	4	2622	2626	2627	6	2017-03-06	2017-03-10	2017-03-11	6.22	6.65
5	5	2661	2665	2666	6	2017-04-14	2017-04-18	2017-04-19	6.48	7.20
6	6	3554	3561	3561	8	2019-09-24	2019-10-01	2019-10-01	5.05	5.93
7	7	3642	3646	3646	5	2019-12-21	2019-12-25	2019-12-25	5.38	6.55
8	8	3686	3687	3690	5	2020-02-03	2020-02-04	2020-02-07	7.06	8.37
9	9	3697	3699	3701	5	2020-02-14	2020-02-16	2020-02-18	6.04	6.61
10	10	3775	3778	3781	7	2020-05-02	2020-05-05	2020-05-08	6.77	8.06



Climate Change and Extreme Weather Events

Heatwaves



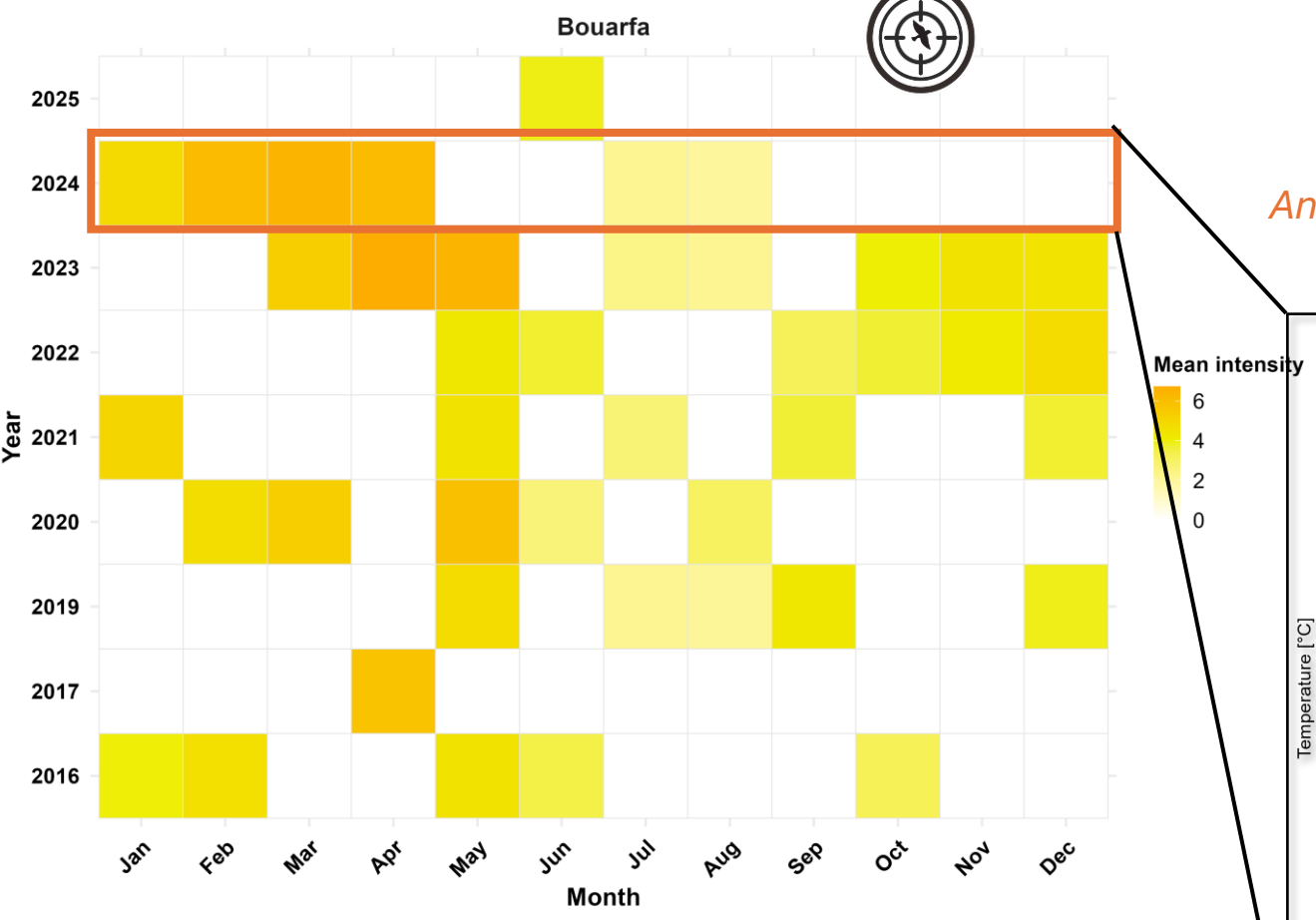
Threshold:
based on 90th –
95th percentile

Baseline :
recommended
over 20-30 years
of data

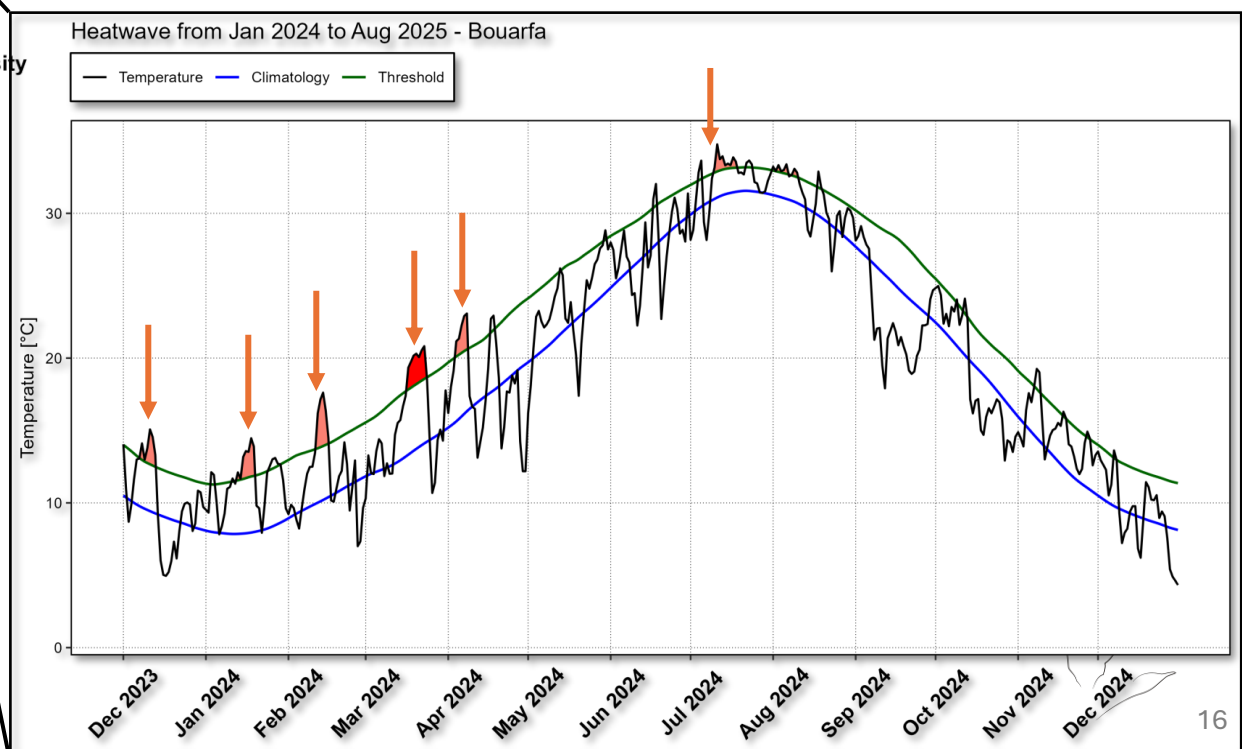


Extreme Weather events -Heatwaves Over Years

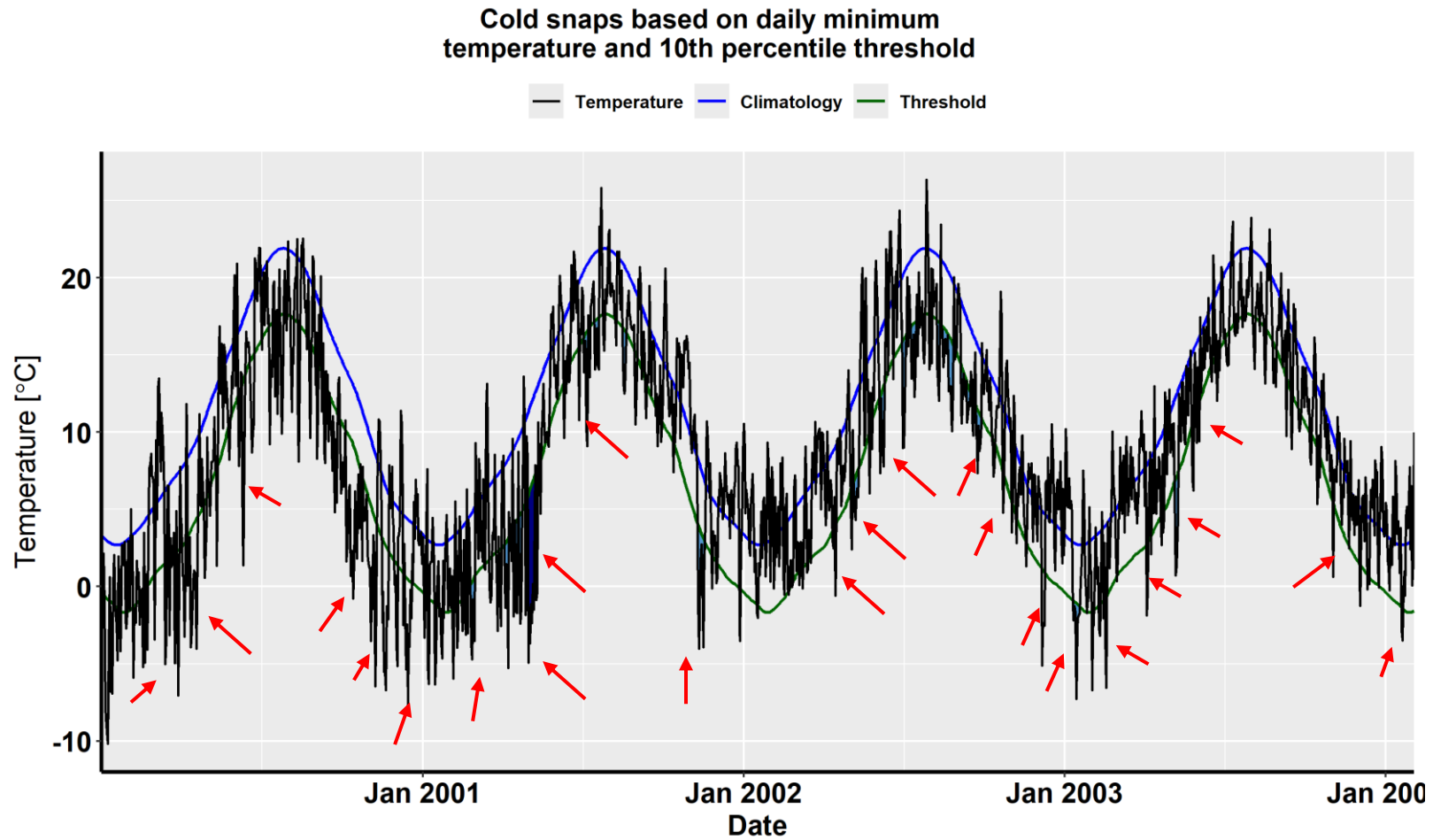
Heatwaves (daily mean temperature and 90th percentile threshold) in Bouarfa



Any spell of ≥ 3 consecutive days with daily temperature above the 90th percentile for that month.

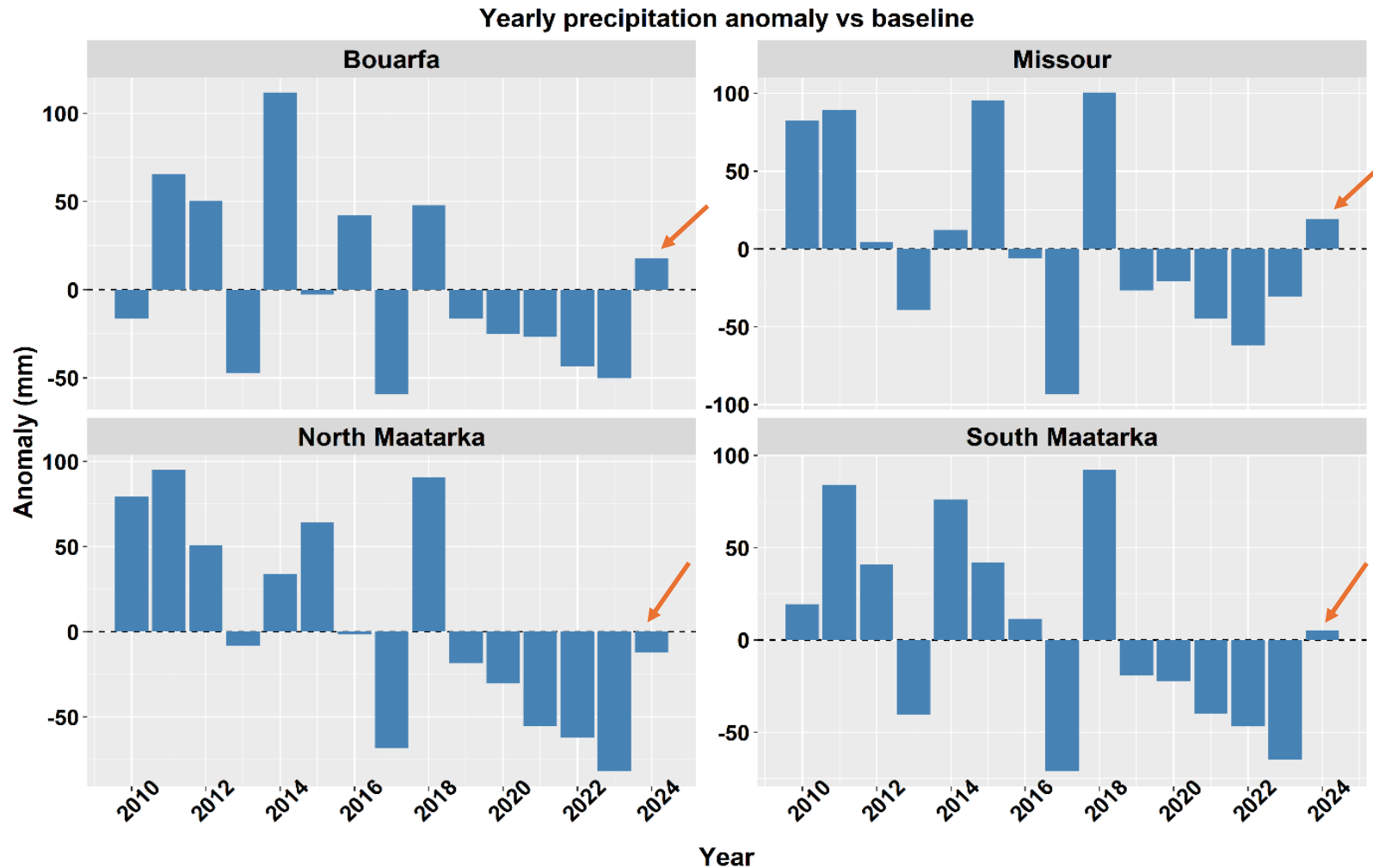


Extreme Weather events - Coldsnaps



Extreme Weather events - Rainfall

Rainfall anomalies = Difference between a year's total rainfall and the **15-year baseline** at that site.



Thank you

